

Vision transformer embedded video anomaly detection using attention driven recurrence

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ABSTRACT

Automated video anomaly detection (VAD) is a challenging task due to its context-dependent and sporadic nature. However, recent deep learning advancements offer promising solutions. In this paper, we propose a novel framework for detecting anomalies in videos by uniquely analyzing spatial and temporal (spatio-temporal) features. We address challenges such as the processing of lengthy videos and the sparse occurrence of anomalies by segmenting and labeling anomalous parts within videos. We employ a modified pre-trained vision transformer for video feature extraction, leveraging its ability to capture complex spatio-temporal patterns and the global context. Additionally, we incorporate a parameter-efficient recurrent model, the Simple Recurrent Unit Plus Plus (SRU++), which processes long sequential video embeddings efficiently by reducing computational costs by ten times compared to traditional methods. To further enhance the multiclass prediction performance, we develop a cluster-based weighting mechanism that assigns weights to classification scores based on feature similarity. We extensively evaluated our approach on three popular datasets — UCF-Crime, RWF-2000, and Smart City CCTV Violence Detection (SCVD) — achieving superior performance compared to state-of-the-art methods, making it well-suited for real-world surveillance applications.

1. Introduction

Surveillance systems are widely deployed in public and private spaces for monitoring and safety purposes. However, manually reviewing hours of footage is slow and inefficient, which makes it imperative to install an automated video anomaly detection (VAD) system. Anomalous events typically represent violations of societal norms or laws and legal standards, contrasting with normal behavior. In our study, we define an anomaly as an event characterized by prolonged rapid movements or drastic changes that lead to chaos, crime, or violence. Accurately detecting such anomalies is challenging due to their long-range contextual dependence, subtle nature of abnormal events, and occurrence over brief moments in extended video sequences [1–3].

Recent advances in deep learning have led to significant improvements in VAD. However, the scarcity of labeled anomalous video data remains a major challenge [4]. Researchers increasingly rely on self- or weakly supervised methods due to the limited availability of annotated datasets [5]. Supervised learning requires appropriately labeled samples, yet most publicly available datasets consist mainly of regular videos with sparse anomaly labels at the video level, making meaningful pattern extraction difficult. Self-supervised methods train only on

normal samples and identify anomalies for unusual patterns [6], while weakly supervised methods use binary classification with ambiguous instance labeling, often applying multiple instance learning (MIL) [4,6]. However, these methods can suffer from mislabeled anomalies, and they often lack generalization to previously unseen anomaly types [6]. Moreover, long video duration with sporadic anomalous events hinders learning complex temporal patterns [5,7].

To enhance automated VAD and its generalization capabilities, it is critical to develop a mechanism that can extract meaningful representations of anomalies and analyze video frames over time efficiently. Moreover, it requires models that can capture long-term dependencies, operate at high recurrence, and support parallel computation for real-time applications. Consider a retail environment or a supershop where a motion sensor is installed. A motion sensor can be triggered by any unusual motion regardless of context, which can lead to many false alarms; for example, a playful child can cause the alarm to go off due to unpredicted movement. A CCTV provides us context and helps us understand when to notify the authority. While human-monitored CCTV provides essential context, manually watching hours of footage is inefficient and prone to human error. To address this, an automated

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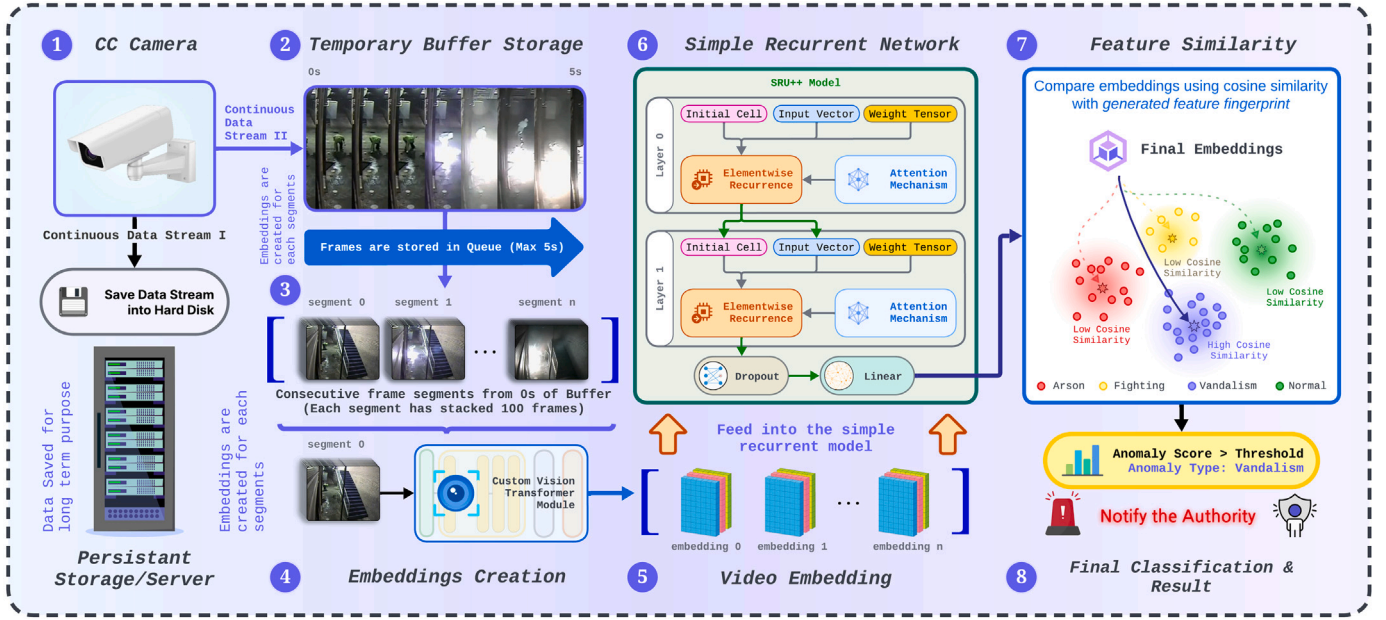


Fig. 1. Abstract diagram of the working mechanism of proposed framework. In a deployed version of the proposed diagram, there are 8 steps that depicts the entire working mechanism. In **step 1**, a CCTV camera will capture continuous video stream which is saved in a persistence storage and in **step 2**, a special high-speed queue-based buffer storage saves 5 s of the continuous video stream. In **step 3**, segments of the video (100 video-frames) are taken from the buffer storage and in **step 4**, each segments are converted into a video embedding by the custom video transformer (converted video embeddings are shown in **step 5**). The video embeddings are then fed into the SRU++ (**step 6**), which captures temporal sequences. The final embeddings from the *Simple Recurrent Unit*, are then compared against feature fingerprints (clusters) using cosine similarity in **step 7** and finally in **step 8**, if anomaly is detected; that is if the anomaly scores exceed certain threshold, the system will notify the authority.

deep learning technique for detecting anomalies in CCTV video can be used, but that has its caveats. One of the primary problems is the deep learning model that works with video data is often slower or lags during inference, especially when they use a vision transformer. This is particularly true for acts of violence, such as a sudden fight erupting between customers. Our proposed method addresses this challenge by using an efficient architecture, which understands the context of the scenario of the environment and does so with significantly lower latency. For example, in the event of a fire, the proposed method should be able to detect the fire before the smoke alarm activates, since a smoke detector requires the fumes to reach its sensor to function, while the proposed efficient model can understand that fire is a hazard and can almost instantly mark it as an anomaly.

In this paper, we propose a comprehensive framework for detecting anomalies, combined with multi-class classification capability. Our method addresses the challenge of long-duration videos such as those in the UCF-Crime dataset, where anomalies occur sparsely. We introduced a temporal segmentation-based labeling strategy that divides lengthy videos into short and more manageable segments, which are cautiously hand-labeled later by humans. This method makes it easier for the model to identify the varied patterns in the categories of events. For spatial feature extraction, we utilize a modified pre-trained Vision Transformer (ViT) model, more specifically, ViT-small. Unlike conventional convolutional neural networks (CNNs), Vision Transformers are able to model global spatial dependencies more effectively due to their self-attention mechanism, which is particularly beneficial in complex video scenes where anomalies may be subtle. Moreover, transformers are known to handle and cancel noisy data by design. In addition, temporal features are modeled using a customized, attention-enhanced Simple Recurrent Unit Plus Plus (SRU++) [8], which is both highly recurrent and designed for efficient parallel computation. SRU++ mainly consists of recurrent units enhanced with attention mechanisms, and its primary benefit over RNN-like architecture is that it can compute up to 10 times faster. Moreover, SRU++ achieves this speed by utilizing the efficient parallel processing capabilities of CUDA architecture, which is crucial for real-time video anomaly detection

where low latency is essential. Furthermore, we incorporated a sliding window system during testing, where long videos are automatically segmented into non-overlapping clips, which are then sequentially passed to the model for evaluation. Lastly, we integrated a cluster-based weighting strategy that enhances multi-class classification accuracy by effectively assigning weights to a particular class to boost the classification score based on feature similarity in the UCF-Crime dataset. We also tested the proposed approach on the RWF-2000 and Smart-City CCTV Violence Detection (SCVD) datasets, where it demonstrated superior performance across these benchmark datasets, including UCF-Crime, by outperforming existing state-of-the-art methods in terms of accuracy and robustness. A general working mechanism is shown in Fig. 1 with 8 steps. The key contributions of this paper include:

1. Developed a new framework for detecting video anomalies that combines a custom vision transformer for spatial feature extraction and SRU++ for temporal feature modeling.
2. For long-duration videos in the UCF-Crime Dataset, we introduced a temporal segmentation-based labeling strategy with a sliding-window mechanism for efficient anomaly detection and classification.
3. Efficiently processes temporal sequences for VAD by utilizing the highly recurrent and parallelized nature of the SRU++ model, which greatly improves training and inference speed.
4. Introduced a cluster-based weighting mechanism that enhances multi-class classification accuracy by leveraging feature similarity with the cluster, particularly for the UCF-Crime dataset.

2. Related work

Anomaly detection has long been a focus in computer vision and machine learning. Abnormal events, both frequent and rare, occur often, and video, being a type of time-series data, requires spatio-temporal feature analysis to detect activities in each frame and the relationships between frames [9,10]. However, multiple solutions have been proposed across supervised, semi-supervised, and unsupervised approaches to enhance anomaly detection.

2.1. Spatio-temporal feature analysis-based methods

Spatial-temporal analysis is widely used in video anomaly detection (VAD), with several studies showing promising results using convolutional neural networks aggregated with LSTMs [11–16]. Petrocchi et al. [17] employed YOLOv4 for object detection, while ResNet-50 was used to extract spatial and motion-related information. Expanding on this, Kim et al. developed a full surveillance system that uses YOLOv5 for human detection and DeepSORT for tracking to create action tubes, which are then fed to a lightweight R(2+1)D network for action recognition, demonstrating robustness in adverse weather conditions [18]. On a similar note, Qasim et al. [13] experimented with different variants of ResNet, such as ResNet-18, ResNet-34, and ResNet-50, followed by a simple recurrent unit that is capable of identifying anomalies within videos by analyzing temporal correlations. Zahid et al. [19] implemented a two-stream efficient network where one stream processes the background and the other processes the difference of other frames. The method uses a pretrained MobileNet for feature extraction and a separable convolutional LSTM that captures long-range spatio-temporal features. To handle the ambiguity of video-level labels, another study employs multiple instance learning (MIL) by considering the lowest anomaly instance score and the highest score in each individual bag with the help of an Inner Bag Loss (IBL) [3]. Similarly, Shao et al. [20] applied MIL to extract multi-scale features, incorporating an attention mechanism to analyze the correlations between positive and negative samples. Furthermore, Abbas et al. [15] incorporates ResNet-50 for feature extraction, which is downsampled via H.265, and every 15 frames are aggregated before being classified using a bidirectional LSTM for temporal modeling. Alternatively, Huszar et al. [21] adopted a different approach by modeling dynamic relationships between actors and objects using the X3D-M3D convolutional network, leveraging the knowledge learned by a pre-trained action recognition model for efficient and accurate violence detection in surveillance footage. On the other hand, A multimodal framework was introduced by Chen et al. [22] that combines visual and text features, so the semantically rich text generated from video captions can cover the limitations of visual features. Other works have explored different modalities and fusion techniques. The MMVAD system processes both video and audio streams, using a Space-Time Transformer for visual features and a dedicated SpectFormer for audio, which are then synchronized through a Cross-Modality Synchronization module [23]. The MMVAD framework also uses vision and language but introduces a novel Attention-based Feature Fusion (AFF) module and a contrastive learning objective to better separate anomaly classes [6]. Leveraging 3D convolutional networks, Choqueluque et al. [24] generate an action tube using a pre-trained detector, followed by Fast-R-CNN for spatio-temporal feature extraction and classification. Lastly, SSIVD-Net [25], which employs a Silent Super-Image approach to simplify complex video data into more useful representations. This method also applies a novel classifier that combines kernel-based methods with residual learning to handle large and complex datasets.

2.2. Reconstruction and prediction-based methods

Reconstruction-based methods have gained popularity under the assumption that anomalies cannot be effectively reconstructed when the model is trained solely on normal data. These approaches, also referred to as one-class classification, learn a model of normalcy from regular events and identify anomalies as deviations from that learned state [5]. Typically, they adopt deep autoencoders to capture normal patterns and use reconstruction error as the primary indicator of an anomaly [4]. However, this approach can be prone to false positives, as a rare but normal event may be misclassified as anomalous simply because it was not seen during training [5,6]. Hasan et al. [26] proposed a couple of autoencoder-based methods for learning regular motion patterns in videos, where one method uses traditional spatio-temporal

features with a fully connected autoencoder, and the other consists of a completely convolutional autoencoder for end-to-end learning. Similarly, Zhao et al. [27] developed a spatio-temporal autoencoder (STAE) model using a 3D convolution for feature extraction and an autoencoder with weighted loss to improve reconstruction accuracy, while Kumaran et al. [28] proposed a hybrid architecture combining a convolutional neural network with a variational autoencoder (CNN-VAE). Moreover, Gong et al. [29] developed Memory-augmented Deep AutoEncoder (MemAE) that combines a deep autoencoder with a memory module to store prototypical data during training and retrieves relevant information while testing that helps to improve poor reconstruction. Lai et al. [30] and Lee et al. [31] designed frameworks, while one leveraged reconstruction loss as spatial attention within an end-to-end model, and another introduced a self-supervised architecture using three contextual prediction streams. On the other hand, Consistency-Aware High-Level Feature Extractor (CAFE) uses a student-teacher architecture to prioritize high-level semantics, and a consistency-aware scheme allows low-level differences to be ignored temporarily [32]. Deng et al. [33] designed a Prior Knowledge Guided Network (PKG-Net) framework by using an auto-encoder network for frame prediction, leveraging a block-wise knowledge distillation technique. In addition, STATE takes advantage of pre-trained object detection models to extract object patches from video frames and generates spatio-temporal information that is passed through a convolutional autoencoder for motion and pixel-level reconstruction [34]. Zhu et al. [35] use a temporal augmented network that consists of an autoencoder, which enables the consideration of motion-aware features alongside spatial and temporal information for anomaly detection. Additionally, an MIL ranking model with attention blocks was used to differentiate between anomalous and non-anomalous video segments by comparing their scores.

2.3. Transformer and attention-based methods

Transformers have been adapted for VAD due to their ability to capture long-term dependencies and spatio-temporal correlations in video sequences; for example, combined frameworks using CNN and Vision Transformers are used where CNN extracts the spatial features, and the transformer learns the temporal relationships [36]. Lee et al. [1] employed a Vision Transformer (ViT) for pixel-level prediction, using three contextual streams for frame reconstruction combined with optical flow reconstruction. Moreover, Bajgoti et al. [37] incorporated a Swin Transformer alongside a generative adversarial network (GAN)-based 3D autoencoder to predict the future frame using a 2D decoder. Similarly, Deshpande et al. [38] also utilized the Swin transformer to generate frame-level anomaly scores, capturing the temporal long- and short-range dependencies. Paulraj et al. also utilized the Swin Transformer as an effective spatial feature extractor, which they combined with a Hybrid Temporal Adaptive Module (HTAM) composed of global self-attention and C-LSTM to model temporal patterns [4]. Other architectures like the Space-Time Transformer have also been employed to efficiently capture both spatial and temporal features from video tubelets [23]. TransUNet and TransAnomaly [39,40] combine vision Transformers with U-Net for the VAD task. Another study incorporated GCN with a discriminative function to filter normal segments and their features from anomalous videos using a contrastive attention module [41]. TIAN by Hannah et al. [42] applies a cross similarity to aggregate input image features globally for video interpolation, which were eventually refined by an image attention module. More recently, large pre-trained Vision-Language Models (VLMs) have been adapted for VAD. The AnomalyCLIP framework is built on top of CLIP, using an Axial Transformer to model temporal dependencies that are absent in the static image features and manipulating the latent space to distinguish between anomaly types [5]. Additionally, Li et al. [2] proposed the Memory-Token Transformer (MMT) that encodes spatio-temporal features by 3D convolution and fuses the semantic tokens of the videos. Li et al. [43] employed two frameworks; one used

the memory addressing module along with an attention scheme, and another introduced a novel loss function for multiple instance learning. Song et al. [44] introduced an Ada-Net, or adversarial attention-based autoencoder network, combined with a graph neural network (GAN) where the decoder is attached with an attention module to selectively focus on informative features.

2.4. LLM-based unsupervised methods

Recent studies in video anomaly detection have increasingly integrated Large Language Models (LLMs) and Vision-Language Models (VLMs) to enhance generalization, reduce training needs, and provide interpretability. Previously mentioned MMVAD, generates text captions from video snippets and uses a dedicated attention-based fusion module to combine them with visual features, followed by a contrastive learning objective to better separate normal and abnormal events across different domains like drone and CCTV footage [6]. LAVAD by Zanella et al. [45] introduced the first ever training-free VAD pipeline using VLM-generated captions and LLM-based anomaly scoring. Moreover, VisionGPT [46] expanded on this breakthrough by enabling real-time, interactive anomaly detection for accessible navigation using YOLO-World and prompt-engineered LLMs. Holmes-VAD by Zhang et al. [47] further improved explainability through an instruction-tuned multimodal dataset. Complementing these, Lv et al. [48] proposed VAD-LLaMA, a multi-stage framework that combines long-term context modeling and instruction tuning for textual justifications, while Shao et al. [49] presented EventVAD, a training-free event-level anomaly detector using dynamic spatiotemporal graphs and hierarchical prompting with MLLMs. Lastly, AnomalyRuler by Yang et al. [50] introduces a rule-based, one-class VAD framework that learns logical anomaly rules from a few normal examples and applies them via LLM reasoning.

Although existing research has made major advancements regarding video anomaly detection, several gaps remain. The literature that relies on methods like CNNs for spatial analysis and LSTMs for temporal modeling, which often struggle to capture long-range dependencies in lengthy videos where anomalous events are sparse. Moreover, unsupervised or reconstruction-based models can be sensitive to noise and may not generalize well. Even advanced methods, including transformers and LLM-based techniques, can be computationally demanding, while our framework integrates the speed of parallel-processed SRU++ with the efficiency of ViT-Small for high performance. Furthermore, weakly supervised methods work with weak labels, where a single label for a long video provides noisy and imprecise temporal information that hinders the model's ability to learn from an actual event. Our method directly confronts this limitation by introducing a temporal segmentation strategy in which we effectively manage long-duration videos and make sparse anomalies more tractable. This approach provides the model with clean, high-quality data and drastically improves its performance. Finally, most of the research focuses on binary anomaly detection without dealing with the more complex and practical task of multi-class classification of different anomaly types, whereas our proposed framework also shows its robust performance on multi-class classification tasks.

3. Methodology

This section presents the proposed video anomaly detection (VAD) framework. The section consists of the dataset description, data preprocessing, feature extraction module, temporal modeling module, and the classification module. Fig. 2 illustrates an overview of our methodology. The rest of the section elaborates on this.

Table 1

Benchmark dataset description. This table provides a detailed overview of the three benchmark datasets commonly used in video anomaly detection studies, highlighting key characteristics such as size and clip length in seconds. Moreover, the table lists the types of anomalies each dataset contains.

Dataset	Total videos	Clip length (s)	Anomalies
UCF-Crime	1900	60–600	Abuse, arrest, arson, assault, accident, burglary, explosion, fighting, robbery, shooting, stealing, shoplifting, vandalism
RWF-2000	2000	5	Fighting, robbery, explosion, shooting, blood, assault
SCVD	484	3–12	Fighting, assault, shooting, weaponized action

3.1. Datasets

The researcher has developed several benchmark datasets to capture diverse types of video anomalies, facilitating extensive research in video anomaly detection. However, these datasets exhibit considerable variation in terms of their anomalous content. For example, datasets like CUHK Avenue, UCSD Pedestrian, and ShanghaiTech focus on non-violent anomalies such as moving in the wrong direction, jumping, pushing, climbing, throwing objects, and individuals with bikes, carts, or wheelchairs. In contrast, other datasets, including Hockey Fights, Movie Fights, and XD-Violence, contain violent scenarios but often represent fictional or staged events, as they are drawn from movies or controlled environments rather than real-world scenarios. However, in this work, the targeted anomalies are characterized by violence, harmfulness, abrupt, high-magnitude motion, and significant environmental changes, such as those found in accidents, robberies, explosions, and other real-world societal crime in public spaces based on surveillance footage. Among the available resources, the UCF-Crime [51] dataset is one of the most comprehensive and appropriate in this regard, containing 1900 long, untrimmed surveillance videos across 13 distinct crime types, including abuse, arson, assault, arrest, road accidents, burglary, explosions, fighting, robbery, shooting, stealing, shoplifting, and vandalism, with an extra normal class; in total, 14 classes. Furthermore, we worked with RWF-2000 [52] and SCVD [25], two smaller datasets containing 2000 and 484 videos of real violence captured by surveillance cameras, respectively. These datasets align with our focus on detecting anomalous activities; therefore, we used these datasets for training and evaluating our model (see Table 1).

3.2. Data preprocessing

We performed extensive preprocessing given the characteristics of the datasets, especially the lengthy videos in UCF-Crime, where anomalies are brief. Our preprocessing steps included segmenting the videos into fixed-length clips, frame reduction, labeling, data augmentation, and finally partitioning the datasets into training and testing sets. The details of the preprocessing steps are as follows, and an overview of the dataset preprocessing is given in Table 2.

3.2.1. Handling data imbalance

The UCF-Crime dataset contains fourteen classes; thirteen of them are anomaly classes, with the remaining one being a normal class. Among the anomaly classes, we excluded the arrest and shoplifting classes from the dataset due to their potential ambiguity for anomaly detection. Apprehending criminals (arrests) is a necessary security measure, not an anomaly. Similarly, shoplifting often involves subtle actions that may vary depending on the context, making it a challenging class for models to learn definitively or may contribute to the higher

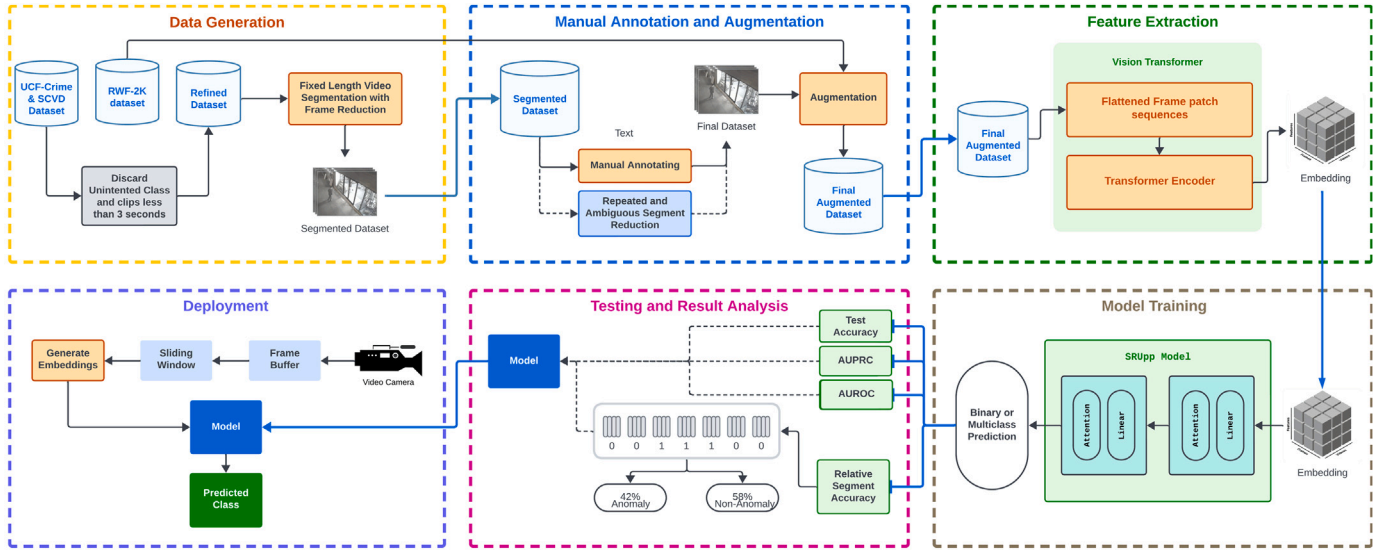


Fig. 2. Overview of the proposed framework. The figure is divided into six parts. In the **Data Generation** part, we generate data for the model by applying the segmentation on the raw datasets into fixed-length clips after discarding unintended classes and videos shorter than a specific duration. The **Manual Annotation and Augmentation** step involves manually labeling these segments, followed by a refinement process that removes repeated or ambiguous segments. Data augmentation is then applied to the refined dataset to increase the size. In the **Feature Extraction** stage, a vision transformer (ViT) processes the video frames by converting them into sequences of flattened patches and feeding them through a transformer encoder to generate high-dimensional spatio-temporal feature embeddings. The **Model Training** phase utilizes these embeddings to train the SRU++ model, which is designed for either binary or multiclass classification of anomalous events. During **Testing and Result Analysis**, the trained model's performance is evaluated on the test set using standard metrics such as accuracy, AUROC, AUPRC and relative segment accuracy (designed for evaluating full length video). Finally, the **Deployment** stage illustrates the real-world application where live video is processed via a sliding window, feature embeddings are generated, and the model predicts the class of the activity in real-time.

false negative rate. The UCF-Crime dataset comprises 1900 videos, split equally with 950 from normal classes and 950 from anomalous classes. Excluding the classes left us with 750 videos from the anomalous class, while the normal class remained unchanged.

We performed statistical analysis on the UCF-Crime dataset, which showed a high skewness in both of the normal and anomalous classes. Our analysis revealed that the anomalous video duration spans from 3.47 s to around 79 min, with a mean of 113.60 and a median of 70.52 s, which indicated positive skewness in the class. Furthermore, the standard deviation (261.44 s) is extremely high compared to the mean, indicating a high variance in the class. Analyzing the interquartile range (IQR) and using the upper fence formula, we found that the statistical outlier threshold is 246 s. Therefore, we deleted all the videos that were longer than 246 s from both of the classes, which left us with 691 videos in the anomalous class and 769 videos in the normal class. We performed similar statistical analysis to handle the data imbalance in the RWF-2000 and SCVD datasets.

3.2.2. Segmentation and frame reduction

We segmented the videos from the UCF-Crime dataset into fixed-length, ten-second video clips. To ensure uniformity in segment length, we discarded the remaining segments that could not cover ten seconds. Furthermore, we excluded videos shorter than ten seconds from the original dataset. In the UCF-Crime dataset, the videos are recorded at 30 frames per second (fps). Therefore, for ten-second segments, it has a total of 300 fps. Following the segmentation, we reduced the number of frames to 100 frames per segment by sampling every third frame. Moreover, the SCVD dataset contains short-length videos ranging from 3 to 11 s, and most of the videos were within 5 to 6 s. Therefore, we split the videos into 5-s segments. Similar to UCF-Crime, we discarded the remaining segments and reduced the number of fps from 30 to 10. However, in the case of the RWF-2000 dataset, the videos were already in 5-s segments, so no further segmentation was required except the frame reduction by one-third. Lastly, we removed all repetitive segments generated by the segmentation technique, preserving only the distinct segments from each scene and angle. This process resulted in

a disproportionately large number of non-anomalous segments; therefore, we proportionally sampled normal data equal to the anomalous data for a balanced dataset consisting of 3500.

Following segmentation and frame reduction, segments were manually labeled and cross-checked through meticulous observation and review. As we selected ten anomalous classes, all of these data were labeled as anomalous for the binary class classification task. Similarly, for the RWF-2000 and SCVD datasets, we labeled the segments as anomalous and non-anomalous, respectively, for binary classification. On the other hand, to enable multiclass classification on the UCF-Crime dataset, we applied distinct labels to each anomalous class.

Subsequently, we performed data augmentation to increase the dataset size and improve the model's generalization capability. We applied random horizontal flipping and random Gaussian noise to the video segments. This augmentation process effectively doubled the dataset size, resulting in a total of 7000 segments for UCF-Crime, 4000 for RWF-2000, and 968 segments for SCVD, preserving the labels. The augmented segments were then divided into training and testing sets by an 80:20 split.

3.3. Vision transformer-based feature extraction module

A specialized class of transformer models known as vision transformers revolutionized the processing of images or videos in computer vision. Unlike convolutional neural networks (CNNs) that focus on local features, vision transformers are capable of capturing long-range dependencies and global context within images, which makes this model effective for understanding complex patterns and relationships in video data. For our model, we use `vit_small_patch16_224` model to generate spatial feature embeddings from video frames. This particular model generates smaller patch size and resolution, that provide a good balance between faster training-inference and computational efficiency.

The ViT model processes an image by breaking it into multiple non-overlapping 16×16 patches using a 16 by 16 kernel. These input patches are then serialized into a vector and linearly projected into patch embeddings. To extract spatial information from the patch embeddings, a learnable positional embedding is added to each patch

Table 2
Dataset statistics comparison.

Metric	UCF-Crime	RWF-2000	SCVD
<i>Video characteristics (Anomaly class)</i>			
Initial videos	1900	2000	484
Video length range (s)	3.2–4730.00	5 (fixed)	3.60–12.90
Mean video length (s)	113.6	5	7.49
Median video length (s)	62.67	5	7.20
<i>Data processing</i>			
Outlier threshold of length (s)	246	–	–
Leftover videos after outlier removal ^a	691	2000	360
Segmentation length (s)	10	5	5
Total segments	6519	2000	791
Leftover segments after removal ^b	3500	2000	500
<i>Final dataset split</i>			
Total segments after augmentation	7000	4000	1000
Training video segments	5600	3200	800
Testing video segments	1400	800	200

^a Both of the outlier videos that crossed the threshold and the videos that were too short (less than 10 s) were removed.

^b All the repetitive and redundant video segments caused by the stationary CCTV camera were removed.

In the table, (s) denotes seconds.

embeddings. The transformer encoder consists of a layer normalization; a multi-head attention that learns different aspects of the relationships between patches; and a feed-forward neural network (MLP block). This MLP block contains two linear layers separated by a GeLU activation function that allows the model to capture more complex patterns and introduce non-linearity. In addition, the encoder contains residual connections which helps in feature preservation.

The encoder provides a sequence of transformed patch embeddings, which are subsequently passed into a spatial projection layer outside the encoder to map these feature embeddings into a fixed dimension. Finally, we accumulate these features to create the feature vector of the entire video, which is input of the SRU++ model. Eq. (1) shows the data transformation before passing into the encoder, where E denotes the patch embeddings and P denotes a patch. In the equation, E_j denotes the j_{th} embeddings originated from the i_{th} patch passed through a linear projection layer, which is added with its corresponding positional embedding Pos_i .

$$E_j = \text{LinearProj}(P_i) + Pos_i \quad (1)$$

where $i \in \{1, 2, \dots, n\}$

Eq. (3) represents the output of the first transformer encoder, denoted by T_0 . T_0 is generated from a series of calculations including Multi-Head Attention (MHA), Feed-Forward Network (MLP) and layer normalization. At first, all the patch embeddings ($E_1, E_2, \dots$) are concatenated together into a single tensor which is denoted by E which is then normalized using a *LayerNorm* function and goes through a *MHA*. The output from the *MHA* is then concatenated with the E itself using a residual connection which is denoted by ϕ and represented in Eq. (4).

$$\phi = E + \text{MHA}(\text{LayerNorm}(E)) \quad (2)$$

where $E = \sum_{i=0}^{16} E_i$

Afterwards, the ϕ is again normalized using *LayerNorm* and goes through a *MLP*. Using another residual connection the output of the *MHA* is concatenated again with the ϕ , resulting the final output (T_0) from the encoder layer.

$$T_0 = \phi + \text{MLP}(\text{LayerNorm}(\phi)) \quad (3)$$

In the transformer architecture, the encoder blocks are stacked sequentially, where the input of each block is the output of the previous block. Let, an encoder that generates output T_n , has the input from the previous encoder block output T_{n-1} . Therefore, a transformer comprises of N layer encoder blocks will have the final encoder layer output T_n . Lastly, Eq. (4) shows that the final feature embedding E_f is generated by applying a projection layer to the output of the final encoder T_n . (see Fig. 3).

$$E_f = \text{ProjectionLayer}(T_n) \quad (4)$$

3.4. Simple recurrent network

A simple recurrent network consists of one or more simple recurrent units (SRU). A simple recurrent unit (SRU) [53] is a light and parameter-efficient recurrent neural network created mainly for natural language processing tasks. It can perform recurrent neural network (RNN) gate calculations extremely fast by utilizing the CUDA¹ framework customization, in which a custom novel matrix multiplication system is built using C++ and the CUDA framework, which is called elementwise recurrence. This enables the model to utilize the full parallel processing power of a GPU, making it nearly ten times faster than traditional RNNs in both training and inference. Working with real-time video data requires very low latency and a super-fast model, and SRU does this job perfectly. Moreover, the inclusion of highway connections facilitates direct information propagation across frame embeddings, aiding in understanding the temporal evolution of visual features. An extended version of SRU is introduced called SRU++ [8] which integrates an attention mechanism in every SRU++ unit. This paper has employed the simple recurrent network with two SRU++ units fused. In our proposed framework, spatial features are extracted by a ViT, and then we feed the extracted feature embeddings into the simple recurrent network, which extracts temporal features and performs the classification task.

Fig. 4 shows the calculation done by the SRU++ cells. The SRU++ applies the elementwise recurrence to calculate the four gates: hidden state h_t , forget gate f_t , reset gate r_t and the current cell state c_t . The calculation equations are given below.

$$f_t = \sigma(W_f x_t + v_f \odot c_{t-1} + b_f) \quad (5)$$

$$c_t = f_t \odot c_{t-1} + (1 - f_t) \odot (W_c x_t) \quad (6)$$

$$r_t = \sigma(W_r x_t + v_r \odot c_{t-1} + b_r) \quad (7)$$

$$h_t = r_t \odot c_t + (1 - r_t) \odot x_t \quad (8)$$

3.4.1. Simple recurrent unit plus plus

An SRU++ unit consists of stacked attention-based recurrent neural network cells that can be connected sequentially. Each cell incorporates a self-attention mechanism to capture long-range dependencies in an input sequence. In an SRU++ unit, we feed an input sequence, $x_{(l,b,f_i)}$ and it generates temporal features $t_{(l,b,f_o)}$, where l , b , f_i , and f_o denote sequence length, batch size, input feature vector dimension, and output feature dimension, respectively. Finally, using the temporal feature embeddings, the model performs classification. Inside an SRU++ unit, the first input $x_{(l,b,f_i)}$ is projected to $x'_{(l,b,f'_i)}$, where f' is the projected feature dimension of shape 2048, then it is passed into the sequential

¹ CUDA is a proprietary parallel computing platform and application programming interface that allows software to use certain types of graphics processing units for accelerated general-purpose processing—an approach called general-purpose computing on GPUs.

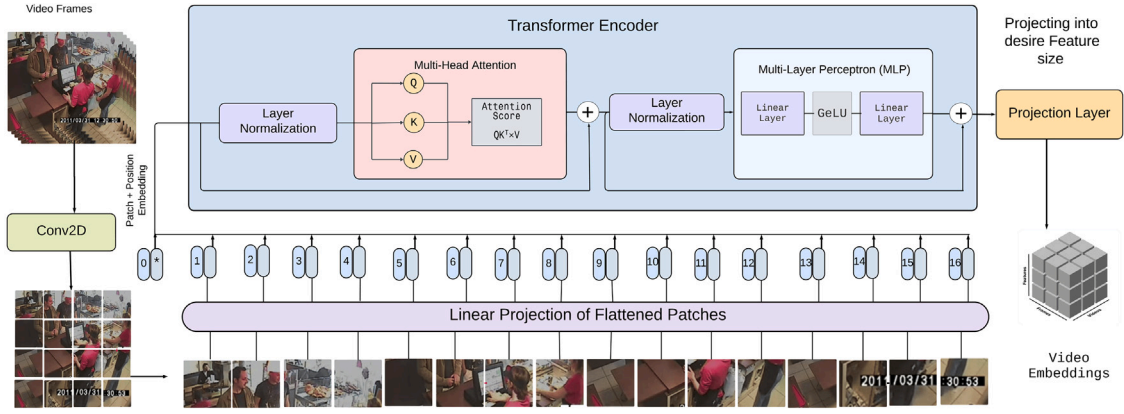


Fig. 3. Feature extraction using vision transformer. An input video frame is divided into patches, which are converted into embeddings with positional information. These are processed by the Transformer Encoder. A custom final projection layer maps the encoder's output into a 2048-dimensional feature vector to create an embedding for the video.

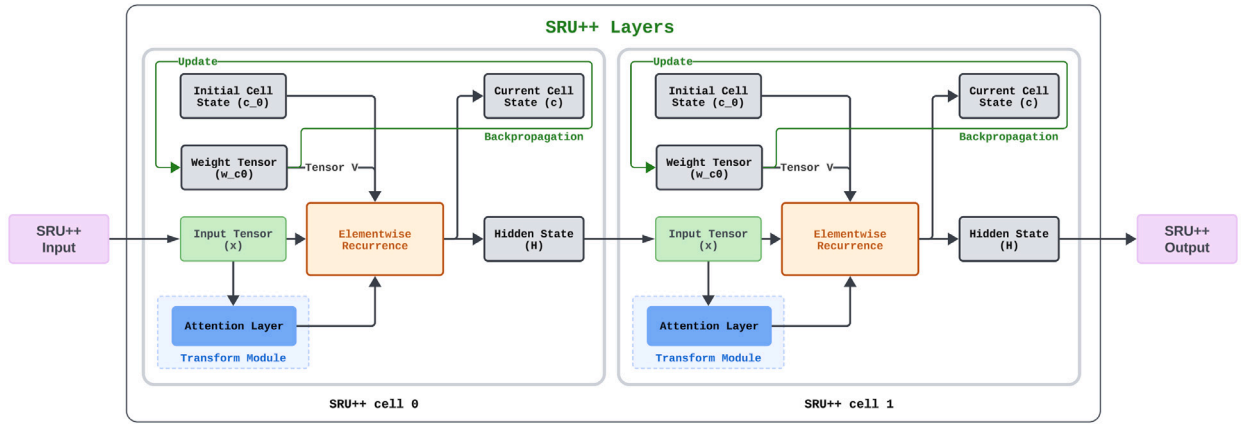


Fig. 4. The architecture of our customized two-layered SRU++ model used for temporal analysis. The input, a sequence of video feature embeddings, is processed by the first SRU++ cell, which uses an attention layer and an elementwise recurrence block to generate a hidden state. This hidden state is then passed to the second cell to capture temporal dependencies. The final output is then used to classify the video segment.

cells of the RNN model, generating a temporal feature $t_{(l,b,f_o)}$ that is expressed in Eq. (9).

$$t_{(l,b,f_o)} = \text{SRU}(x'_{(l,b,f'_i)}) \quad (9)$$

Finally, the temporal feature $t_{(l,b,f_o)}$ is processed by a dropout layer Δ_{dropout} followed by a linear layer $\mathcal{L}$ that maps the features into the class numbers n and predicts the class probability P of the corresponding classes C_n .

$$P(C_n) = \mathcal{L}(\Delta_{\text{dropout}}(t_{(l,b,f_o)})) \quad (10)$$

3.4.2. Multi-head attended simple recurrent cell

The model consists of two cells that are connected sequentially. Each cell generates a hidden state h_i that propagates to the input of the next cell, a cell state c_i . At first, a multi-headed attention mechanism is applied to the projected input $x'_{(l,b,f'_i)}$ and creates attention output $U_{(l,b,f_o)}$. It consists of eight attention heads, which allow the module to attend to eight different aspects, capturing broader and more diverse relationships within the frame sequence. Query Q , key K and value V are extracted from the projected input $x'_{(l,b,f'_i)}$ from two linear layers $\mathcal{L}_Q$ and $\mathcal{L}_{KV}$. Eqs. (11) and (12) depict how Q , K and V are extracted. Each cell has a weight matrix $V_{(2 \times f_o)}$ through backpropagation and a cell state c_0 . The attended simple recurrent cell calculates all the gate values by elementwise recurrence technique using CUDA-level optimization. The elementwise recurrence produces hidden state h_i and cell state c_i , which propagate to the next layer.

$$Q = \mathcal{L}_Q(x'_{(l,b,f'_i)}) = W_Q x'_{(l,b,f'_i)} \quad (11)$$

$$K, V = \mathcal{L}_{KV}(x'_{(l,b,f'_i)}) = W_K x'_{(l,b,f'_i)}, W_V x'_{(l,b,f'_i)} \quad (12)$$

In the next step, eight attention heads are created from the Q , K and V with their weight matrices W_Q , W_K and W_V . Each head contains query, Q_i , a key, K_i and value V_i where i is the attention head index. For each head, we scale the query matrix Q by a factor of $\frac{1}{\sqrt{d_k}}$, where d_k is the feature dimension of the key K matrix, which is later multiplied by the transposed key matrix K^T to calculate the attention scores (logits). A softmax activation function and dropout are applied to these scores to obtain the attention weights, followed by matrix multiplication with the value matrix V , resulting in the attention output U .

$$U = \Delta_{\text{dropout}} \left(\text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) \right) V \quad (13)$$

3.5. Feature cluster weighing for multiclass classification

Our approach also includes multiclass classification for the UCF-Crime dataset since it has various categories of annotated classes. In binary class classification, all anomalous classes are considered as anomaly classes, which is easier for the model to classify compared to multiclass classification. However, for multiclass classification, the model has to learn all characteristic patterns and has a high chance of misclassifying. We incorporated a cluster weighing module to effectively weight the classification score for each test sample using a clustering technique. Initially, we clustered the training data using their spatio-temporal features, and the cluster centers were refined at each

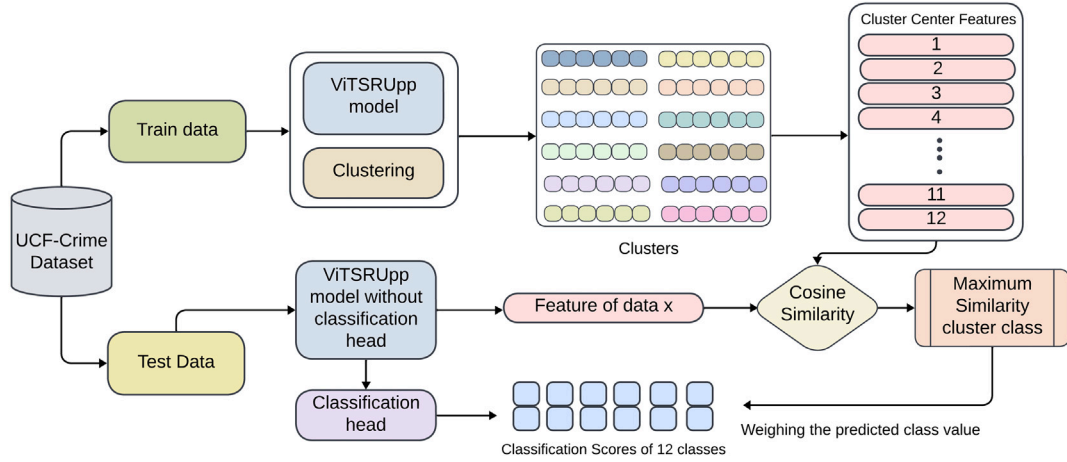


Fig. 5. Train data is clustered at the time of training while refining the cluster center throughout the training. On the other hand, the feature of the test data is compared with the separated cluster centers using cosine similarity, and the maximum similarity cluster class is weighted with the classification scores.

training step; therefore, the final cluster centers were derived at the end of the training. In the test phase, for each data point, the spatio-temporal feature was extracted using the feature extraction module before passing to the classification head. Their features were compared to the features of each cluster center using the cosine similarity, which is given in Eq. (13). The cluster with the highest cosine similarity was considered. Therefore, when a test data point aligns with a specific cluster, we assign a weight to the prediction score of that particular class, given in Eq. (14). This weight boosts the confidence level of the test data point belonging to that particular class. By leveraging this technique, we prioritize their influence on the final classification decision.

$$\text{Cosine Similarity}(\mathbf{d}_i, \mathbf{c}_j) = \frac{\mathbf{d}_i \cdot \mathbf{c}_j}{\|\mathbf{d}_i\| \|\mathbf{c}_j\|} \quad (14)$$

$$p[c_x] = p[c_x] \times w, \quad \text{where } c_x = \text{sim}(\mathbf{d}_i, \mathbf{c}_j)_{\max} \quad (15)$$

where $\mathbf{d}_i$, $\mathbf{c}_j$ denotes the data points and clusters, respectively, and $\mathbf{c}_j$ represents the cluster centers. Moreover, $p[c_x]$ denotes the prediction scores generated by the model for each class. The class c_x is assigned a weight for which the test data gets the maximum similarity (see Fig. 5).

3.6. Learning algorithm

Algorithm 1 presents the complete training algorithm for our automated video anomaly detection framework, which integrates feature extraction using ViT, SRU++ for temporal modeling, and cluster-based weighting for enhanced multiclass classification performance.

The training process consists of five main phases: (1) **Feature Extraction** using pre-trained ViT to convert video frames into 2048-dimensional embeddings, (2) **Temporal Modeling** with SRU++ to capture sequential dependencies, (3) **Classification** through linear layers with optional cluster-based weighting (only for multiclass classification), (4) **Loss Computation** combining cross-entropy and L2 regularization, and (5) **Parameter Updates** using Adam optimizer with backpropagation.

The algorithm incorporates several key optimizations, such as the cluster centers being updated periodically to adapt to evolving feature distributions, dropout regularization preventing overfitting, and early stopping based on validation performance ensuring optimal convergence.

Algorithm 1 Training Algorithm for Video Anomaly Detection

Require: Training dataset $D = \{(V_i, y_i)\}_{i=1}^N$, pre-trained ViT model
Require: Hyperparameters: learning rate α , epochs E , number of clusters K
Ensure: Trained model parameters θ^*

- 1: Initialize ViT from pre-trained weights, SRU++ parameters randomly
- 2: Initialize K cluster centers using k-means on sample features
- 3: **for** epoch $e = 1$ to E **do**
- 4: **for** each batch $\{(V_j, y_j)\}$ in D **do**
- 5: // **Forward Pass**
- 6: Extract frame embeddings: $\mathbf{X}_j = \text{ViT}(V_j)$
- 7: Model temporal dependencies: $\mathbf{H}_j = \text{SRU}++(\mathbf{X}_j)$
- 8: Compute raw predictions: $\mathbf{P}_{\text{raw},j} = \text{Linear}(\mathbf{H}_j)$
- 9: // **Cluster-based Weighting**
- 10: Apply feature similarity weighting: $\mathbf{P}_j = \text{Weight}(\mathbf{P}_{\text{raw},j}, \mathbf{H}_j)$
- 11: // **Optimization**
- 12: Compute loss: $\mathcal{L} = \text{CrossEntropy}(\mathbf{P}_j, y_j) + \lambda \|\theta\|_2$
- 13: Update parameters: $\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}$
- 14: **end for**
- 15: Update cluster centers every 10 epochs
- 16: Validate and apply early stopping if needed
- 17: **end for**
- 18: **return** Optimized parameters θ^*

4. Experimental evaluations

To comprehensively assess the performance of our proposed video anomaly detection framework, we measured several standard evaluation metrics commonly used in binary and multiclass classification tasks, such as accuracy, area under ROC curve (AUC) score, and precision-recall curve (PRC) score.

Accuracy measures the proportion of correctly classified samples out of the total number of samples. The area under the ROC curve (AUC) shows the true positive rate against the false positive rate at various threshold settings. AUC provides an aggregate measure of performance across all classification thresholds, with values ranging from 0 to 1, where 1 indicates perfect classification. The area under the precision-recall curve (PRC) provides a more informative metric for imbalanced datasets where the positive class (anomalies) is rare. We report macro-averaged metrics for multiclass classification on the UCF-Crime dataset, ensuring that all classes receive equal weight, regardless of their frequency.

These metrics provide a comprehensive evaluation of our model's ability to distinguish between different types of anomalies and normal behavior across various surveillance scenarios. Most of the VAD work reported the aforementioned metrics; therefore, we also reported these metrics for better comparison. The following subsections offer more detailed information about the implementation, result analysis, and ablation studies.

4.1. Implementation details

In this study experiment, frames were reduced in an even interval manner, meaning that every second frame was taken. For example, if a video has 30 frames per second, 15 of them will be reduced for every second. For a 10-s video segment, we collected 150 frames, and for a 5-s video segment, we collected 75 frames. Subsequently, the frames were resized to $224 \times 224 \times 3$ of height, width, and channel, respectively, preparing the feature extraction module. We used a Python library, *timm*, to create a pre-trained model of a vision transformer that sequentially processes the frames using CUDA, returning a 2048 size embedding for each frame. At the end of the phase, we get a shape-independent feature embedding of shape (7000, 150, 2048) for the UCF-Crime, (4000, 75, 2048) for the RWF-2000, and (1000, 75, 2048) for the SCVD dataset, representing the total videos, total frames, and embedding size of the frames. It is worth noting that for UCF-Crime dataset, the context length of the SRU++ model is 150 frames, whereas for RWF-2000 and SCVD datasets, it is 75 frames. After splitting the dataset into a train test, the training subset is fed into the SRU++ model sequentially in $b_n = 8$ batches. The SRU++ model is configured with two cells, empirically selected to be optimal. The model processes a 2048-dimensional feature vector, generating a hidden state h_i and a cell state c_i where c_i is discarded due to its internal usage in recurrent gate calculation. A dropout layer of 0.2, followed by a fully connected layer, is applied in the hidden state h_i that classifies the video into different classes. Eight attention heads are used with a scaling factor of $\frac{1}{\sqrt{h_d}}$ where h_d is the attention head dimension. For training, the *crossEntropy* loss function and the *Adam* optimizer are used with the model's internal parameters. In loss calculation, L2 regularization with strength $\lambda = 1 \times 10^{-5}$ is used to encourage the model to learn smaller parameter values. All hyperparameters were empirically chosen for optimal performance.

4.2. Results on benchmark datasets

All the datasets were trained on different numbers of epochs with different parameter tuning; 100 yielded the optimal results for UCF-Crime and RWF-2000; 50 for the SCVD dataset. Primarily, our model works as a binary classifier since the initial priority is to identify any violence in a place regardless of its category. Afterward we employed the same architecture for multiclass classification only for the UCF-Crime, given that the videos were annotated as 12 different classes. It has shown a robust and promising result in terms of both AUC and PRC. Table 3 shows the train and test accuracies, AUC and PRC scores for the optimal number of epochs, along with their required runtime for ResNet-50 with SRU++ and ViT with SRU++, respectively. Moreover, Table 4 represents the Latency and memory size comparison of the model. This table compares the latency of the models in our proposed framework. We calculated the latency of ViT and SRU++ model on CPU and GPU hardware. Moreover, we calculated the raw model size and compressed model size on the disk space.

Since our evaluation also includes multiclass classification on the UCF-crime dataset, we provide the ROC and PR curves for the multiclass classification in Fig. 6. The ROC and PRC scores for all classes ranges from 0.93–1.00 and 0.82–0.98, respectively, demonstrating the effectiveness of our cluster-based weighting mechanism and further highlighting the robustness of the overall framework.

4.3. Quantitative evaluation

To verify the effectiveness of our method, we compared with other state-of-the-art VAD methods by measuring the accuracy and AUC score. Table 5 presents the accuracy scores of binary classification of different methods along with our method experimented on three benchmark datasets. The methods include [13,20–22,48,54–57] for the UCF-Crime dataset; [19,21,24,25,58–62] for the RWF-2000 dataset; and lastly, [25,63–65] for the SCVD dataset. Among the existing works, some evaluated their methods using the AUC score, others used accuracy, and a few reported both. Our model consistently outperformed these benchmark approaches and achieved state-of-the-art performance across all three datasets. However, [59,64] obtained higher accuracy score than ours for the RWF-2000 and SCVD datasets, respectively. Most state-of-the-art models on the UCF-Crime dataset are weakly supervised, as the dataset provides only video-level labels for long-duration videos containing sparse and sporadic anomalous events. In contrast, the other two datasets are more concise and contain clearly defined anomalous events, therefore, the methods are supervised. In our method, the lengthy videos of UCF-Crime were split into smaller chunk that helped our model identify the context better and faster, demonstrating the superior capability of vision transformers and SRU in understanding violence in supervised manner. Additionally, we have also demonstrated comparisons with other methods in terms of hyperparameters and computational resources in Table 7.

4.4. Anomaly visualization

To show the visual explanation of our model's prediction, we generated a representation of the performance of our model across six distinct classes from the UCF-Crime dataset. We propagate full-length videos through the end-to-end model. In return, it generates segments using a sliding window approach and makes predictions on each segment. Anomalous segments typically yielded high anomaly scores, which caused the prediction graphs to show distinct peaks, whereas normal video segments produced lower scores, resulting in a flattened graph at the bottom. For example, if a normal video displays a shop with few customers in a calm environment, it generates less than a 0.5 anomaly score for most segments. Although some segments that exceed 0.5 produce false negatives in segment-level predictions, these do not affect video-level predictions because, in our relative accuracy measurement system, the overall decision is measured based on the majority of correct segment predictions. Similarly, videos of explosions showed clear peaks in segments where flames were present. In road accident videos, although most segments contained regular traffic, the model successfully identified and scored highly the few segments containing accidents. Although the anomaly scores for the assault segments were not as high as other anomalous classes, the model effectively captured these patterns with a confidence score exceeding 0.5. However, the model accurately detected the burglary and fighting segments, which contain the true anomalies, with high confidence scores. Fig. 7 illustrates the prediction performance across these six categories using four representative segments for each.

4.5. Ablation study

We performed our ablation study for different combinations of our model on three datasets. Initially, we focused on robust feature extraction using convolutional neural networks. Feature extraction was initially performed using basic Convolutional Neural Networks (CNNs) augmented with a weighted attention module, which is input to the SRU module for comprehensive spatio-temporal analysis and classification. To enhance the result, we further trained with SRU++, which includes a projected self-attention mechanism on its layers that generated a better result. In the next stage, a pre-trained ResNet-50 was

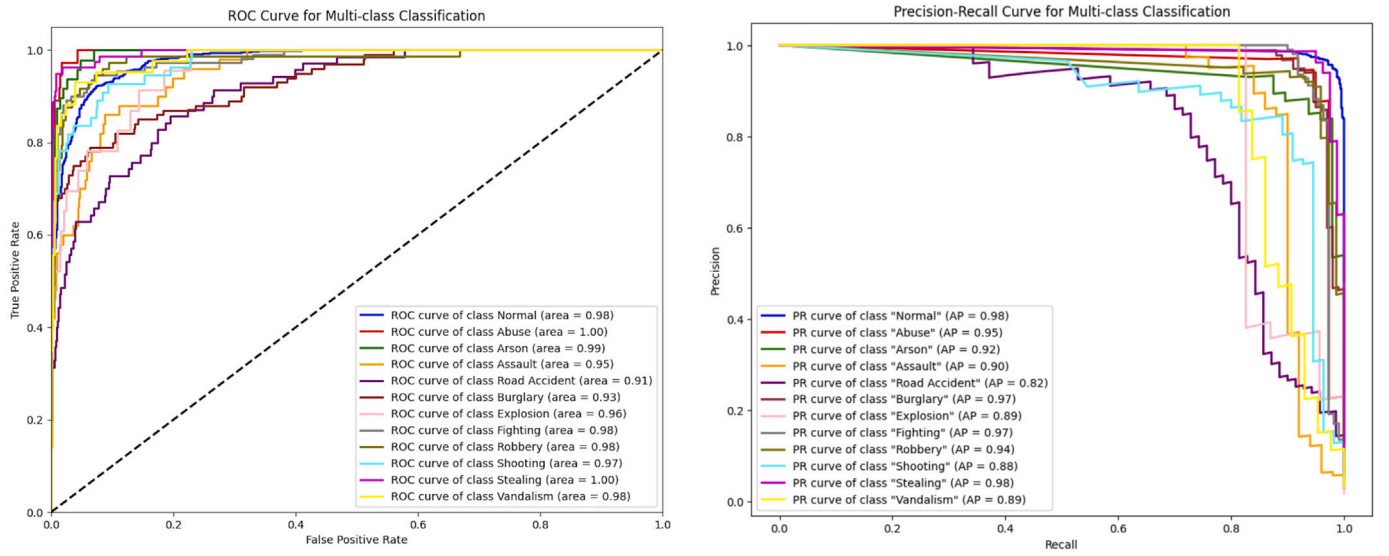


Fig. 6. ROC and PRC scores of 12 classes on multi-class classification for UCF-Crime dataset.

Table 3

Experimental results and a comparison between feature extractor ResNet-50 and ViT with SRU++. The first three row represents results when ResNet-50 feature extractor is used during training, and the last three column represents results when ViT is used. With a longer training time, model with ViT achieves more score than with ResNet-50.

Dataset	Train Acc.	Test Acc.	AUC score	PRC score	Epoch	Training time (min)
UCF-Crime	96.31%	93.65%	0.95	0.94	100	232
RWF-2000	93.43%	90.38%	0.93	0.92	100	115
SCVD	93.27%	89.80%	0.92	0.94	50	53
UCF-Crime	97.58%	95.14%	0.96	0.94	100	245
RWF-2000	93.7%	92.88%	0.95	0.94	100	119
SCVD	96.77%	93.73%	0.93	0.95	50	58

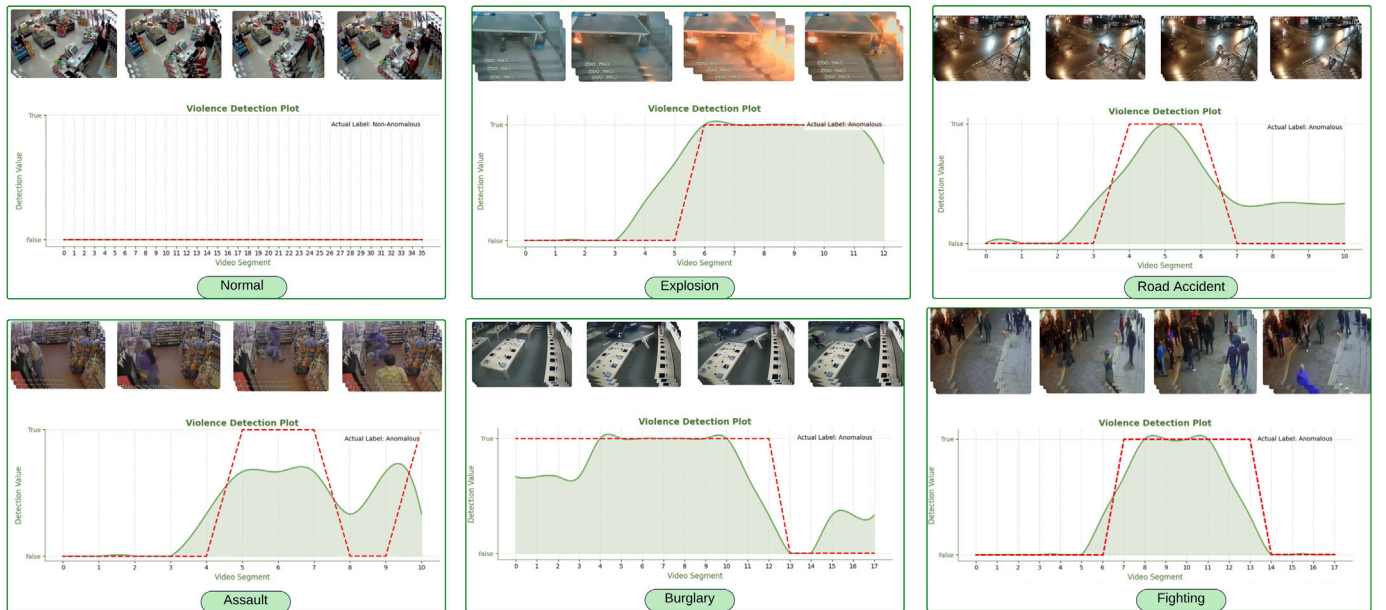


Fig. 7. Visual examples of prediction results for 6 different classes of UCF-Crime dataset, including Normal, Explosion, Road Accident, Assault, Burglary, and Fight. Each class has a different number of segments based on its length. Here we have shown 4 sequential representative segments on top of the graphs. Our model is able to generate accurate detection by generating high anomaly scores of its corresponding segments.

Table 4
Model latency and memory size comparison.

Device	ViT	SRU++ (binary)	SRU++ (Multiclass)
CPU latency ^a	13738.5 ms	383.20 ms	461.49 ms
GPU latency (CUDA) ^a	5149.50 ms	1.35 ms	1.38 ms
Model size	85.76 MB	96.08 MB	96.08 MB
Model size (int8 quantized) ^b	22.78 MB	24.07 MB	26.01 MB

MB denotes Megabytes and ms denotes Milliseconds.

^a We used Kaggle platform CPU and GPU (P100) to calculate the latency. It denotes an average time a model takes to process a batch of videos. Latency is calculated by averaging the process time of 167 batches of videos, where a batch represents 8 videos.

^b For model compression, we applied int8 quantization technique.

incorporated for feature extraction, followed by SRU and SRU++ in succession. The obtained results in this stage were impressive, demonstrating comparability with state-of-the-art models. We initially employed a pre-trained ResNet for frame-level feature extraction. Subsequently, a Vision Transformer (ViT) was incorporated to enable more comprehensive video-level spatio-temporal feature learning, as the vision transformer divides the image into fixed-size patches instead of processing the entire image at once. These patches are added with a positional embedding, helping in understanding the position embeddings of the patches during training. Moreover, the transformer encoder consisting of self-attention makes it possible to capture relationships between different patches. In the end, we used both SRU and SRU++ for training and temporal reasoning. Both of these architectural combinations significantly outperformed all previously tested configurations. Table 6 summarizes the binary classification accuracies for all evaluated combinations across the datasets.

5. Limitations and future work

While our framework achieves high accuracy in detecting anomalies in videos, its architecture has limitations. The models used in this pipeline require high computation power. The Table 4, clearly shows that the vision transformer (ViT), at 22.1 million parameters, takes about 5.15 s to generate embeddings of 8 videos, each containing 150 frames. It takes thrice that time on the CPU, which leads to a bottleneck. Although GPUs could mitigate latency, their cost makes them unfeasible for edge setups. Vision transformers can run on low-cost edge devices such as Raspberry Pi, which is perfect for a CCTV setup; however, achieving low latency would be a major challenge. Real-time video anomaly detection requires ultra-low latency; otherwise, it may overflow the buffer storage, where the live footage is stored in a queue to be processed. The second model in our framework is the SRU++, which has 2 layers of RNN layers, each layer containing one attention layer with 8 attention heads. Despite its primary design for GPUs, the model can also operate solely in a CPU environment. We tested the model's capability by running it in both CPU and GPU environments; as shown in Table 4, SRU++ performs almost instantly in the GPU environment, while its performance in the CPU environment is average. Combining the two heavy models, the cumulative latency can lead to major performance impact in a real-world scenario. Furthermore, in this research, we are using a pre-trained ViT model as a feature extractor, which can extract less meaningful features in terms of video anomalies. A pre-trained vision transformer may struggle to extract features that highlight subtle and sudden changes or elements like fire, which are often indicative of an anomaly. Additionally, a limited dataset is not sufficient to generalize all the anomaly scenarios, which may result in false positives or true negatives.

To address these issues, the research can be progressed further. Implementing the ViT in a C++ codebase can significantly reduce latency and memory usage, given the computationally expensive nature of the framework. SRU++ has the potential to run better on the CPU if it can be reimplemented using C++ so that it can execute commands

directly on the CPU using multiple thread pools; however, this proposal needs more research. Moreover, model quantization will help reduce the computation further but with a trade-off between latency and accuracy. Exporting the framework in ONNX format will reduce the model size, making it more edge-friendly. Additionally, future studies can include introducing a newly trained feature extractor specifically for anomaly detection and a more robust clustering technique for multiclass classifications. Lastly, as real-world anomaly contains scenarios that can never be in the training data due to unpredictable nature, the model can be constantly fine-tuned using the data gathered by the system in real-time upon permission.

6. Conclusion

In conclusion, we have developed a novel automated video anomaly detection (VAD) framework for efficient violence detection in surveillance videos, which utilizes a custom vision transformer for deep spatial feature extraction and integrates a high-recurrence SRU++ module for efficient and robust anomaly classification. The proposed approach achieves competitive performance compared to existing models for anomaly detection and accurately classifies anomalous events. Experiments on three benchmark datasets demonstrate that training our base model with segmentation techniques significantly contributes to the efficiency and context understanding in complex scenes. Furthermore, we proposed a sliding window mechanism during inference using a long video without segmentation. However, challenges such as false negatives in low-resolution footage and false positives in visually chaotic environments persist. Additionally, the computational demands of the vision transformer are a consideration. In future work, we plan to experiment with more lightweight vision transformers, apply model quantization and optimization, and reduce overhead through techniques such as weight and bias pruning so that the model can be deployed on resource-constrained edge devices. Additionally, the work can be extended to investigate predictive modeling approaches for the early identification of potentially harmful behaviors.

Data availability

Codes are available at <https://github.com/mariamuna04/VAD>.

CRediT authorship contribution statement

Ummay Maria Muna: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Shantha Biswas:** Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Syed Abu Ammar Muhammad Zarif:** Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Philip Jefferson Deori:** Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Tauseef Tajwar:** Writing – review & editing, Validation, Methodology, Conceptualization. **Swakkhar Shatabda:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 5

Quantitative Performance comparison of different methods (leftmost column) on UCF-Crime, RWF-2000, and SCVD datasets with accuracy (ACC) and area under the ROC curve (AUC) metrics. For each dataset, there is two columns of metrics that shows the consecutive reported score of the methods.

Author	Training strategy	UCF-Crime		RWF-2000		SCVD	
		ACC (%)	AUC (%)	ACC (%)	AUC (%)	ACC (%)	AUC (%)
Chen et al. [22]	Weakly supervised	84.9	91.2	–	–	–	–
Shao et al. [20]	Weakly supervised	85.1	91.5	–	–	–	–
Zhang et al. [54]	Weakly supervised	86.20	89.78	–	–	–	–
Karim et al. [55]	Weakly supervised	–	87.48	–	–	–	–
LV et al. [48]	Weakly supervised	–	88.13	–	–	–	–
Gao et al. [56]	Weakly supervised	–	91.34	–	–	–	–
Sharif et al. [57]	Weakly supervised	–	88.97	–	–	–	–
Qasim et al. [13]	Supervised	91.64	94.1	–	–	–	–
Huszar et al. [21]	Supervised	95.0	95.8	88.5	92.1	–	–
David et al. [24]	Supervised	–	–	88.71	90.6	–	–
Islam et al. [19]	Supervised	–	–	89.75	91.7	–	–
Hamid et al. [62]	Supervised	–	–	90.4	93.2	–	–
Su et al. [58]	Supervised	–	–	90.0	–	–	–
Senadeera et al. [59]	Supervised	–	–	94.0	–	–	–
Garcia et al. [60]	Supervised	–	–	90.25	–	–	–
Sato et al. [61]	Unsupervised	–	–	90.3	–	–	–
Aremu et al. [25]	Supervised	–	–	90.3	92.9	86.6	91.4
Tu et al. [63]	Supervised	–	–	–	–	90.7	–
Wankhade et al. [64]	Supervised	–	–	–	–	96.4	–
Li et al. [65]	Supervised	–	–	–	–	90.74	–
Ours	Supervised	95.14	96.0	92.88	95.0	93.73	93.0

Note:

ACC and AUC denote Accuracy and Area Under the Receiver Operating Characteristic Curve score.

‘–’ represents values were not explicitly mentioned in the paper.

Table 6

Accuracy comparison of different versions of proposed model architectures and feature extraction techniques (CNN, ResNet-50, ViT) combined with SRU and SRU++ on UCF-Crime, RWF-2000, and SCVD datasets.

Dataset	CNN+SRU	CNN+(SRU++)	ResNet-50+SRU	ResNet-50+(SRU++)	ViT+SRU	ViT+(SRU++)
UCF-Crime	85.28%	87.79%	91.24%	93.65%	94.57%	95.14%
RWF-2000	83.63%	84.95%	88.10%	90.38%	91.23%	92.88%
SCVD	82.18%	84.46%	87.53%	89.80%	91.50%	93.73%

Table 7

Parameter comparison between all other literature and the proposed method.

Parameters	Chen [22]	Shao [21] TL model	Shao [21] FT model	Zahidul [19]	Aremu [25]	Abbas [15]	Ours ViT	Ours ResNet-50
Input size ^a	2048	2048	65 536	150 528	50 176	50 176	2048	2048
FE ^b	ID3	X3D-M	X3D-M	MobileNetv2	CNN ^h	ResNet-50	ViT	ResNet-50
Classes	1	2	2	2	3	–	12	12
GPU ⁱ	V100	1080Ti	1080Ti	P100	RTX6000	MX450	P100	P100
LR ^c	0.001	0.001	0.001	0.0004	0.0001	0.00001	0.001	0.001
Optimizer	Adam	Adagrad	Adagrad	AMSGrad	SGD	Adam	Adam	Adam
Batch size	32	30	4	4	–	16	8	8
Epoch	1000	50	50	150	50	100	100	100
Dropout	0.7	0.1	0.1	–	–	0.7	0.2	0.2
LF ^d	RTFM ^e	BCE ^f	BCE	BCE	–	–	BCE	BCE
TP ^g	~28M	~4M	~2M	~0.3M	<i>j</i>	–	~35.5M	~38.6M

– (Hyphen ‘–’) represents values that were not explicitly mentioned in the paper.

M denotes Million.

^a Input size refers to the dimension of the features that the model takes. Number represents flatten feature size.

^b FE represents Feature Extractor.

^c LR represents Learning Rate.

^d LF represents Loss Function.

^e RTFM represents a custom loss function incorporating sparsity, SigmoidMAELoss, SigmoidCrossEntropyLoss with L1 penalty.

^f BCE represents Binary Cross Entropy Loss.

^g TP represents the number of total parameters.

^h Authors have used a modified CNN which includes residual networks.

ⁱ NVIDIA GPUs are used.

^j Value depends on the model. The authors created several models, with parameter sizes ranging from 1.8M to 8M.

References

- [1] Lee J, Nam W-J, Lee S-W. Multi-contextual predictions with vision transformer for video anomaly detection. In: 2022 26th international conference on pattern recognition. ICPR, IEEE; 2022, p. 1012–8.
- [2] Li Y, Song X, Xu T, Feng Z. Memory-token transformer for unsupervised video anomaly detection. In: 2022 26th international conference on pattern recognition. ICPR, IEEE; 2022, p. 3325–32.
- [3] Zhao M, Liu Y, Liu J, Zeng X. Exploiting spatial-temporal correlations for video anomaly detection. In: 2022 26th international conference on pattern recognition. ICPR, IEEE; 2022, p. 1727–33.

- [4] Paulraj Shalmiya, Vairavasundaram Subramaniaswamy. Transformer-enabled weakly supervised abnormal event detection in intelligent video surveillance systems. *Eng Appl Artif Intell* 2025;139:109496, URL <https://www.sciencedirect.com/science/article/pii/S0952197624016543>.
- [5] Zanella Luca, Liberatori Benedetta, Menapace Willi, Poiesi Fabio, Wang Yiming, Ricci Elisa. Delving into CLIP latent space for video anomaly recognition. *Comput Vis Image Underst* 2024;249:104163, URL <https://www.sciencedirect.com/science/article/pii/S1077314224002443>.
- [6] Biswas Debojyoti, Tesic Jelena. MMVAD: A vision-language model for cross-domain video anomaly detection with contrastive learning and scale-adaptive frame segmentation. *Expert Syst Appl* 2025;285:127857, URL <https://www.sciencedirect.com/science/article/pii/S0957417425014794>.
- [7] Ren J, Xia F, Liu Y, Lee I. Deep video anomaly detection: Opportunities and challenges. In: 2021 international conference on data mining workshops. ICDMW, IEEE; 2021, p. 959–66.
- [8] Lei T. When attention meets fast recurrence: Training language models with reduced compute. 2021, arXiv:2102.12459.
- [9] Chang Yunpeng, Tu Zhigang, Xie Wei, Luo Bin, Zhang Shifu, Sui Haigang, Yuan Junsong. Video anomaly detection with spatio-temporal dissociation. *Pattern Recognit* 2022;122:108213.
- [10] Wang Yang, Liu Tianying, Zhou Jiaogen, Guan Jihong. Video anomaly detection based on spatio-temporal relationships among objects. *Neurocomputing* 2023;532:141–51.
- [11] Sudhakaran S, Lanz O. Learning to detect violent videos using convolutional long short-term memory. In: 2017 14th IEEE international conference on advanced video and signal based surveillance. AVSS, IEEE; 2017, p. 1–6.
- [12] Vosta S, Yow K-C. A CNN-RNN combined structure for real-world violence detection in surveillance cameras. *Appl Sci* 2022;12(3):1021.
- [13] Qasim M, Verdu E. Video anomaly detection system using deep convolutional and recurrent models. *Results Eng* 2023;18:101026.
- [14] Zhang J, Qing L, Miao J. Temporal convolutional network with complementary inner bag loss for weakly supervised anomaly detection. In: 2019 IEEE international conference on image processing. ICIP, IEEE; 2019, p. 4030–4.
- [15] Abbas ZK, Al-Ani AA. Anomaly detection in surveillance videos based on H265 and deep learning. *Int J Adv Technol Eng Explor* 2022;9(92):910.
- [16] Ullah Farman Ullah Muhammad, Muhammad Khan, Haq Ihsan Ullah, Khan Nadeem, Heidari Ali Akbar, Baik Sung Wook, de Albuquerque V, Hugo C. AI-assisted edge vision for violence detection in IoT-based industrial surveillance networks. *IEEE Trans Ind Inform* 2021;18(8):5359–70.
- [17] Petrocchi S, Giorgi G, Cimino M G. A real-time deep learning approach for real-world video anomaly detection. In: Proceedings of the 16th international conference on availability, reliability and security. 2021, p. 1–9.
- [18] Kim Hyungmin, Jeon Hobeom, Kim Dohyung, Kim Jaehong. Elevating urban surveillance: A deep CCTV monitoring system for detection of anomalous events via human action recognition. *Sustain Cities Soc* 2024;114:105793, URL <https://www.sciencedirect.com/science/article/pii/S2210670724006176>.
- [19] Islam Z, Rukonuzzaman M, Ahmed R, Kabir MH, Farazi M. Efficient two-stream network for violence detection using separable convolutional LSTM. In: 2021 international joint conference on neural networks. IJCNN, IEEE; 2021, p. 1–8.
- [20] Shao W, Xiao R, Rajapaksha P, Wang M, Crespi N, Luo Z, Minerva R. Video anomaly detection with NTCN-ML: A novel TCN for multi-instance learning. *Pattern Recognit* 2023;143:109765.
- [21] Huszar VD, Adhikarla VK, Négyesi I, Krasznay C. Toward fast and accurate violence detection for automated video surveillance applications. *IEEE Access* 2023;11:18772–93.
- [22] Chen W, Ma KT, Yew ZJ, Hur M, Khoo DA. TEVAD: Improved video anomaly detection with captions. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. IEEE; 2023, p. 5549–59.
- [23] Paulraj Shalmiya, Vairavasundaram Subramaniaswamy. M2VAD: Multiview multimodality transformer-based weakly supervised video anomaly detection. *Image Vis Comput* 2024;149:105139, URL <https://www.sciencedirect.com/science/article/pii/S0262885624002439>.
- [24] Choqueluque-Roman D, Camara-Chavez G. Weakly supervised violence detection in surveillance video. *Sensors* 2022;22(12):4502.
- [25] Aremu T, Zhiyuan L, Alameeri R, Khan M, Saddik AE. SSIVD-net: A novel salient super image classification and detection technique for weaponized violence. In: Science and information conference. Springer; 2024, p. 16–35.
- [26] Hasan Mahmudul, Choi Jonghyun, Neumann Jan, Roy-Chowdhury Amit K, Davis Larry S. Learning temporal regularity in video sequences. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2016, p. 733–42.
- [27] Zhao Yiru, Deng Bing, Shen Chen, Liu Yao, Lu Hongtao, Hua Xian-Sheng. Spatio-temporal autoencoder for video anomaly detection. In: Proceedings of the 25th ACM international conference on multimedia. 2017, p. 1933–41.
- [28] Santhosh Kelathodi Kumaran, Dogra Debi Prosad, Roy Partha Pratim, Mitra Adway. Vehicular trajectory classification and traffic anomaly detection in videos using a hybrid CNN-VAE architecture. *IEEE Trans Intell Transp Syst* 2021;23(8):11891–902.
- [29] Gong Dong, Liu Lingqiao, Le Vuong, Saha Budhaditya, Mansour Moussa Reda, Venkatesh Svetha, Hengel Anton van den. Memorizing normality to detect anomaly: Memory-augmented deep autoencoder for unsupervised anomaly detection. In: Proceedings of the IEEE/CVF international conference on computer vision. 2019, p. 1705–14.
- [30] Lai Y, Liu R, Han Y. Video anomaly detection via predictive autoencoder with gradient-based attention. In: 2020 IEEE international conference on multimedia and expo. ICME, IEEE; 2020.
- [31] Lee J-Y, Nam W-J, Lee S-W. Multi-contextual predictions with vision transformer for video anomaly detection. 2022, arXiv:2206.08568.
- [32] Yu G, Wang S, Cai Z, Liu X, Wu C. Effective video abnormal event detection by learning a consistency-aware high-level feature extractor. In: Proceedings of the 30th ACM international conference on multimedia. MM'22, ACM; 2022.
- [33] Deng Z, Chen D, Deng S. Prior knowledge guided network for video anomaly detection. 2023, arXiv:2309.01682.
- [34] Wang Y, Qin C, Bai Y, Xu Y, Ma X, Fu Y. Making reconstruction-based method great again for video anomaly detection. 2023, arXiv:2301.12048.
- [35] Zhu Y, Newsam S. Motion-aware feature for improved video anomaly detection. 2019, arXiv:1907.10211.
- [36] Ullah W, Hussain T, Ullah FUM, Lee MY, Baik SW. TransCNN: Hybrid CNN and transformer mechanism for surveillance anomaly detection. *Eng Appl Artif Intell* 2023;123:106173.
- [37] Bajgoti A, Gupta R, Balaji P, Dwivedi R, Siwach M, Gupta D. SwinAnomaly: Real-time video anomaly detection using video swin transformer and SORT. *IEEE Access* 2023.
- [38] Deshpande K, Punn NS, Sonbhadra SK, Agarwal S. Anomaly detection in surveillance videos using transformer-based attention model. In: International conference on neural information processing. Springer; 2022, p. 199–211.
- [39] Chen Jieneng, Lu Yongyi, Yu Qihang, Luo Xiangde, Adeli Ehsan, Wang Yan, Lu Le, Yuille Alan L, Zhou Yuyin. Transunet: Transformers make strong encoders for medical image segmentation. 2021, arXiv preprint arXiv:2102.04306.
- [40] Yuan Hongchun, Cai Zhenyu, Zhou Hui, Wang Yue, Chen Xiangzhi. Transanomaly: Video anomaly detection using video vision transformer. *IEEE Access* 2021;9:123977–86.
- [41] Chang S, Li Y, Shen S, Feng J, Zhou Z. Contrastive attention for video anomaly detection. *IEEE Trans Multimed* 2021;24:4067–76.
- [42] Kim HH, Yu S, Yuan S, Tomasi C. Cross-attention transformer for video interpolation. In: Proceedings of the Asian conference on computer vision. 2022, p. 320–37.
- [43] Li Q, Yang R, Xiao F, Bhanu B, Zhang F. Attention-based anomaly detection in multi-view surveillance videos. *Knowl-Based Syst* 2022;252:109348.
- [44] Song Hao, Sun Che, Wu Xinxiao, Chen Mei, Jia Yunde. Learning normal patterns via adversarial attention-based autoencoder for abnormal event detection in videos. *IEEE Trans Multimed* 2019;22(8):2138–48.
- [45] Zanella Luca, Menapace Willi, Mancini Massimiliano, Wang Yiming, Ricci Elisa. Harnessing large language models for training-free video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2024, p. 18527–36.
- [46] Wang Hao, Qin Jiayou, Bastola Ashish, Chen Xiwen, Suchanek John, Gong Zihao, Razi Abolfazl. Visiongpt: Llm-assisted real-time anomaly detection for safe visual navigation. 2024, arXiv preprint arXiv:2403.12415.
- [47] Zhang Huaxin, Xu Xiaohao, Wang Xiang, Zuo Jialong, Han Chuchu, Huang Xiaonan, Gao Changxin, Wang Yuehuan, Sang Nong. Holmes-vad: Towards unbiased and explainable video anomaly detection via multi-modal llm. 2024, arXiv preprint arXiv:2406.12235.
- [48] Lv Hui, Sun Qianru. Video anomaly detection and explanation via large language models. 2024, arXiv preprint arXiv:2401.05702.
- [49] Shao Yihua, He Haojin, Li Sijie, Chen Siyu, Long Xinwei, Zeng Fanhu, Fan Yuxuan, Zhang Muyang, Yan Ziyang, Ma Ao, et al. Eventvad: Training-free event-aware video anomaly detection. 2025, arXiv preprint arXiv:2504.13092.
- [50] Yang Yuchen, Lee Kwonjoon, Dariush Behzad, Cao Yinzhi, Lo Shao-Yuan. Follow the rules: reasoning for video anomaly detection with large language models. In: European conference on computer vision. Springer; 2024, p. 304–22.
- [51] Sultani W, Chen C, Shah M. Real-world anomaly detection in surveillance videos. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2018, p. 6479–88.
- [52] Cheng M, Cai K, Li M. RWF-2000: an open large scale video database for violence detection. In: 2020 25th international conference on pattern recognition. ICPR, IEEE; 2021, p. 4183–90.
- [53] Lei T, Zhang Y, Wang SI, Dai H, Artzi Y. Simple recurrent units for highly parallelizable recurrence. 2018, arXiv:1709.02755.
- [54] Zhang Yiling, Akdag Erkut, Bondarev Egor, et al. MTFE: multi-timescale feature learning for weakly-supervised anomaly detection in surveillance videos. In: Seventeenth international conference on machine vision. ICMV 2024, Vol. 13517, SPIE; 2025, p. 166–73.
- [55] Karim Hamza, Doshi Keval, Yilmaz Yasin. Real-time weakly supervised video anomaly detection. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision. 2024, p. 6848–56.
- [56] Gao Andrew, Liu Jun. STEAD: Spatio-temporal efficient anomaly detection for time and compute sensitive applications. 2025, arXiv preprint arXiv:2503.07942.

- [57] Sharif Md Haidar, Jiao Lei, Omlin Christian W. CNN-ViT supported weakly-supervised video segment level anomaly detection. *Sensors* 2023;23(18):7734.
- [58] Su Yukun, Lin Guosheng, Wu Qingyao. Improving video violence recognition with human interaction learning on 3D skeleton point clouds. 2023, arXiv preprint [arXiv:2308.13866](https://arxiv.org/abs/2308.13866).
- [59] Senadeera Damith Chamalke, Yang Xiaoyun, Kollias Dimitrios, Slabaugh Gregory. Cue-net: violence detection video analytics with spatial cropping enhanced uniformerv2 and modified efficient additive attention. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2024, p. 4888–97.
- [60] Garcia-Cobo Guillermo, SanMiguel Juan C. Human skeletons and change detection for efficient violence detection in surveillance videos. *Comput Vis Image Underst* 2023;233:103739.
- [61] Sato Fumiaki, Hachiuma Ryo, Sekii Taiki. Prompt-guided zero-shot anomaly action recognition using pretrained deep skeleton features. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2023, p. 6471–80.
- [62] Mohammadi H, Nazerfard E. Video violence recognition and localization using a semi-supervised hard attention model. *Expert Syst Appl* 2023;212:118791.
- [63] Tu YS, Shen YS, Chan YY, Wang L, Chen J. Violent video recognition by using sequential image collage. *Sensors (Basel)* 2024;24(6):1844.
- [64] Wankhade Atharva, Jaiswal Snehal, Tingane Snehal. Violence detection in surveillance videos using artificial intelligence. *Int J Eng Res Modernization (IJERM)* 2024;11(5):9, URL https://www.ijerm.com/download_data/IJERM1105009.pdf.
- [65] Li M, Ren X, Fu H, Li Z, Sun J. ConvMHSA-SCVD: Enhancing smart contract vulnerability detection through a knowledge-driven and data-driven framework. In: *2023 IEEE 34th international symposium on software reliability engineering*. ISSRE, Florence, Italy; 2023, p. 578–89.